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Review

Mapping the Global Nursing Literature on Aromatherapy in Surgery: A Bibliometric Analysis

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A B S T R A C T

Purpose: This study aimed to bibliometrically analyze nursing publications on aromatherapy used in surgery.
Design: The study is a bibliometric analysis.
Methods: The bibliometric analysis study was conducted using the Web of Science database, with the data analyzed up to April 1, 2025. The analysis was performed using the "Biblioshiny" tool from the Bibliometrix R package.
Findings: The bibliometric analysis identified 102 relevant studies on aromatherapy in surgical nursing, published between 2006 and 2025. The highest number of publications occurred in 2021 (n = 15). The most productive journal was *Journal of Perianesthesia Nursing*. Keywords frequently used included "aromatherapy," "anxiety," "pain," "lavender," and "nursing and surgery." The thematic analysis highlighted key research areas such as pain management, anxiety reduction, and the use of lavender oil in surgical settings. The most influential countries in this field were Iran, with notable contributions from journals focused on complementary therapies.
Conclusions: This bibliometric study provides a comprehensive overview of global research trends, focal topics, and thematic evolution concerning the use of aromatherapy in nursing. This bibliometric study provides a comprehensive overview of global research trends, focal topics, and thematic evolution concerning the use of aromatherapy in surgical nursing. The findings may guide future research directions and highlight areas requiring further exploration in evidence-based nursing practice.

Aromatherapy is an approach that has been practiced in many countries for years and holds a significant place among complementary medicine methods. Contributions from ancient civilizations such as the Egyptians, Chinese, and Indians, as well as from Arab cultures, have laid the foundation for the contemporary understanding and current applications of aromatherapy.[1] In ancient Egypt and the Middle East, plant oils were used in embalming, fumigation, and healing. The development of clinical aromatherapy began in France just before World War I, when chemist Maurice Gattefossé treated a gangrenous wound with lavender essential oil. Gattefossé later advocated the use of aromatherapy for infections and war wounds. Physician Jean Valnet and nurse Marguerite Maury followed Gattefossé in Europe, emphasizing the importance and use of essential oils. By the 1930s, the importance of essential oils in preventing infections began to emerge in European and Australian medical literature.[2] Gaining popularity particularly since the 1980s, this method has been widely used in nursing practice to alleviate physical and psychological symptoms.[3] Aromatherapy involves the application of essential oils, which are naturally extracted from flowers and plants, through methods such as massage or inhalation. When administered via inhalation, the olfactory system is stimulated, sending signals to the limbic system, the emotional control center of the brain.[4] Stimulation of the limbic system by aromas leads to the release of neurochemicals that reduce pain and support brain functions, thereby inducing a sense of calmness. Literature indicates that aromatherapy can reduce heart rate, blood pressure, and respiratory rate, while also helping to restore hormonal balance.[5] In aromatherapy massage applications, essential oils are absorbed through the skin into the circulatory system, providing therapeutic benefits such as analgesic effects, muscle relaxation, and reduced anxiety.[6] Due to its noninvasive nature, low cost, ease of application, and lack of chemical side effects, this method has become one of the most used complementary therapies in nursing practices.[7-9]

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The American Nurses Association also emphasizes that this therapeutic approach should be considered an integral component of holistic nursing care.[10] Furthermore, nurses are expected to integrate this method into their professional practices, including the use of essential oils, in accordance with evidence-based practices.[11] Research in the field of nursing has demonstrated that aromatherapy is highly effective in alleviating symptoms such as preoperative anxiety, depression, stress, pain, insomnia, fatigue, nausea, and vomiting. In this context, aromatherapy is increasingly being preferred as an alternative or complementary method to pharmacological treatments.[11-18] Aromatherapy is also widely used as a complementary method in the surgical field, particularly in reducing preoperative anxiety. In a study conducted by Stanley et al[12] on patients undergoing cataract surgery, a significant reduction in anxiety levels was observed in Stanley.[12] Similarly, in another study by Karaman and colleagues,[18] aromatherapy using ginger, lavender, and rose oils was found to be effective in alleviating postoperative nausea and vomiting. Surgical nursing is a specialty that embraces holistic nursing care encompassing the pre-, intra-, and postsurgical periods. The decision to operate before surgery can trigger psychological reactions in the patient, leading to anxiety and stress. Various aromatherapy studies have been conducted to reduce preoperative anxiety and stress and evaluate the effectiveness of these methods.[5,12,19-21] Lavender aromatherapy was administered via inhaler for preoperative anxiety in female patients undergoing elective surgery, and patients who received lavender aromatherapy reported lower anxiety levels.[20] Increased stress and anxiety in surgical patients can lead to physiological responses such as increased postoperative pain and nausea.[19] Various surgical procedures have been performed by applying different essential oils through massage and inhalation to reduce postoperative pain levels.[22-27] The effectiveness of ginger, lavender essential oil, lemon, and mint aromatherapy in relieving postoperative nausea and vomiting has been evaluated.[27-30] Rambod et al[27] evaluated the effect of lemon inhalation aromatherapy on pain, nausea, vomiting, and neurovascular assessment in patients undergoing lower-extremity fracture surgery and showed that aromatherapy reduced pain intensity, postoperative nausea, vomiting, and burning sensation, as well as the frequency of antiemetic drug administration.[27]

Bibliometric analysis is one of the statistical methods applied to enable researchers from various countries and disciplines to access information and analyze it more comprehensively. This analytical method provides a systematic framework for mapping the development of a research field, evaluating research findings through numerical data, offering objective criteria, and identifying new trends in the field.[31] On reviewing the literature, no bibliometric study specifically focusing on the use of aromatherapy in surgery has been identified. This original study aims to determine the focus and thematic trends of research conducted on aromatherapy in surgical contexts. The findings are expected to serve as a guide for future authors working in this field.

Method

Aim

This study aims to examine the research on aromatherapy in the field of surgery as presented in the literature, to identify current trends and focal topics, and to provide guidance for future studies.

Research Questions

-What is the distribution of nursing research on aromatherapy in surgery in terms of countries, journals, and keywords?

-What are the trends, most productive topics, and existing gaps in the nursing research literature related to aromatherapy in surgery?

Screening Strategy

Considering the status of being the most widely accepted database for scientific publication analysis, the Web of Science (Wos) database (https://www.webofscience.com/wos/woscc/basic-search), accessed via the online library of Akdeniz University, was selected as the search engine for this study. Due to the ongoing flow of scientific publications into the WoS database and its continuously updated nature, the data were collected on a single day April 1, 2025, according to the inclusion and exclusion criteria. WoS is recognized as one of the most established and comprehensive citation databases of scholarly publications worldwide. Its preference in scientific evaluation processes, particularly in bibliometric analyses, provides a significant advantage in terms of the scientific validity and reliability of the research being conducted. The WoS database includes studies with high scientific merit that comply with publishing ethics standards and offers a wide range of literature for research conducted across different disciplines. For these reasons, the WoS database was chosen as the data source in the context of this study.[32]

In the database search strategy, MeSH terms and their combinations were used. The following search terms were applied: TS = (“aromatherapy” OR “aromatherapy massage” OR “aromatherapy inhalation” OR “essential oils”) AND (“surgery” OR “operative” OR “general surgery”) AND (“nurse” OR “nursing” OR “nursing staff” OR “nurses”). Quotation marks were placed around each term, and no time restrictions were applied during the literature search. The inclusion criteria were (1) original research articles, systematic reviews, and early- access publications, and (2) articles published in English. The exclusion criteria were (1) conference proceedings, books, book chapters, and editorial materials, and (2) duplicate studies. Based on these keywords, a total of 117 studies were retrieved. After applying the inclusion and exclusion criteria, the final number of studies included was 102.

To construct the bibliometric map, all bibliographic data were exported from the WoS database using the “Save to Other File Formats” option. In this export, plain text format was selected to include the full record and cited references, and the file was saved as “Savedrecs.txt.” Since the search yielded more than 500 entries, articles were downloaded in batches of 500 and then merged into a single file. The bibliometric analysis procedure and implementation guidelines designed by Donthu et al. (2021) were adopted in this study (Figure 1).[33]

Data Analysis

In the data analysis process, the Biblioshiny interface—part of the open-source R programming.[34] environment (version 4.3.3, R Foundation)—was utilized within the Bibliometrix package, a programming language developed specifically for statistical analysis and offers advanced capabilities for visualizing data through graphical representations.[34] Bibliometrix, written in R, facilitates interconnection with other R packages.[34] In this study, bibliometric analysis techniques were conducted using tools available in the R environment. The Biblioshiny interface provided a user-friendly platform to carry out the analysis. Both the R language and the Bibliometrix package are widely recommended for performing in-depth scientific mapping and bibliometric evaluations.

Ethical Considerations

Ethical approval was not required for this study.

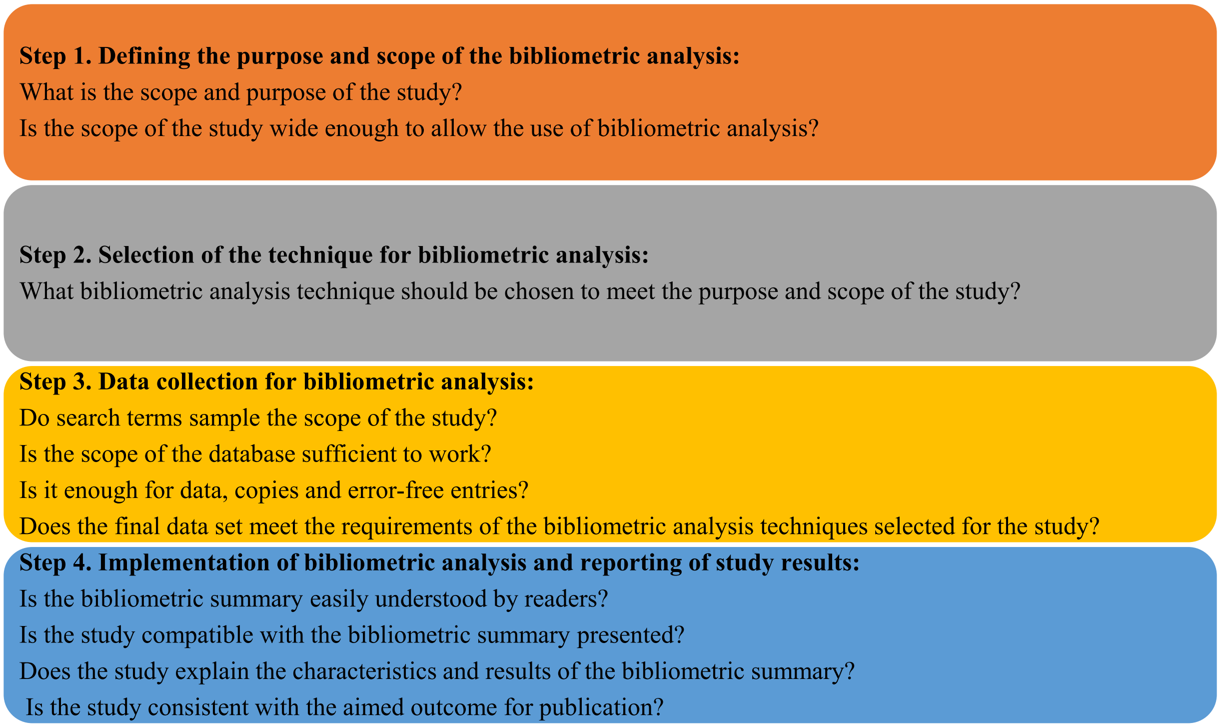

Figure 1. Bibliometric analysis procedure and best practice guidelines.

Results

Main Information About Publications

Table 1 presents general information regarding the characteristics of the publications. The studies in this field were conducted between the years 2006 and 2025. The average time since publication is 5.2 years. A total of 408 authors contributed to the studies. The country of the most prolific author was Iran, while the journal with the highest number of publications on the subject was the *Journal of Perianesthesia Nursing* (Table 1).

Publication Output and Growth Trend

In this section, it was observed that the highest number of articles was published in 2021 (n = 15), followed by 2024 (n = 13), and the third highest in 2020 and 2022 with 12 articles each (Figure 2).

Table 1
Main Information About Publications

Basic Knowledge	Results
Data	
Sources	56
Documents	102
Time space	2006-2025
Document age of average	5.2
Annual growth rate of publications	7.57%
Writers	
Authors	408
Author's keywords	245
References	3216
Document types	
Article	84
Review	18
Top three most published publications	
Journal of Perianesthesia Nursing	19
Explore—The Journal of Science and Healing	7
Holistic Nursing Practice	6
Country of the top three most published authors	
Iran	38
United States	20
Türkiye	9
Search terms	TS = ("aromatherapy" OR "aromatherapy massage" OR "aromatherapy inhalation" OR "essential oils") AND ("surgery" OR "operative" OR "general surgery") AND ("nurse" OR "nursing" OR "nursing staff" OR "nurses")

Figure 2. Publication output and growth trend.

Figure 3. World cloud.

World Cloud, Thematic Map, and Trend Topic Analysis

It was found that 245 author keywords were used in studies on aromatherapy applied in surgery within the field of nursing. A word cloud of the 50 most frequently used author keywords is shown in Figure 3. The word cloud increases in size with the rising frequency of word usage. The most frequently used author keywords were identified as aromatherapy (n = 57), anxiety (n = 31), pain (n = 20), lavender (n = 11), and nursing and surgery (n = 10) (Figure 3).

Figure 4 presents the thematic map of research on aromatherapy applied in surgery in the nursing field. This thematic map focuses on the development and popularity of keywords over the years and is analyzed with centrality on the x axis and density on the y axis. Centrality reflects the strength of a topic's connection and importance in the field, while density shows how deeply and extensively that topic has been developed in research. In this analysis, the number of words is 250, the minimum cluster reference is 5, and there is 1 level per cluster.

Motor themes, characterized by higher density and high centrality, are located in the upper right quadrant. Here, the themes consist of two words, namely "pain," "health," and "mood," which are among the most studied topics. The upper left quadrant, which contains niche themes, features lower centrality and higher density. These themes are less connected to the main topic. Niche themes in this quadrant include "prevention," "cancer," and "cognitive-behavioral therapy." The lower

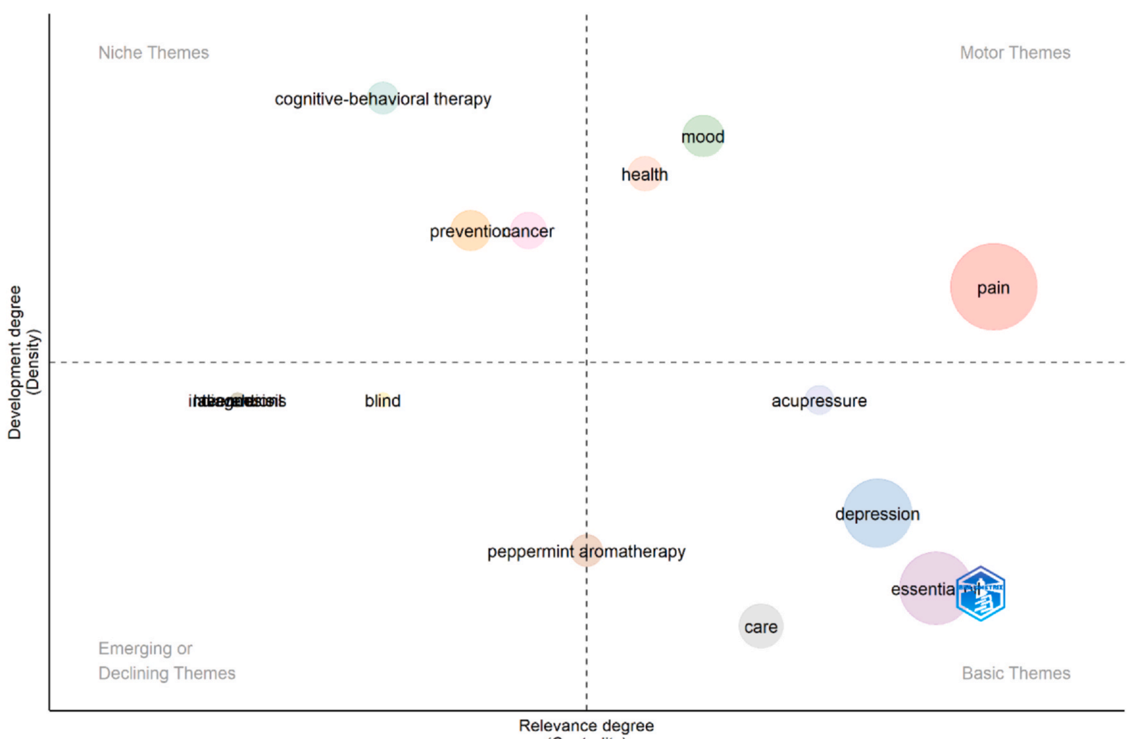

Figure 4. Thematic map.

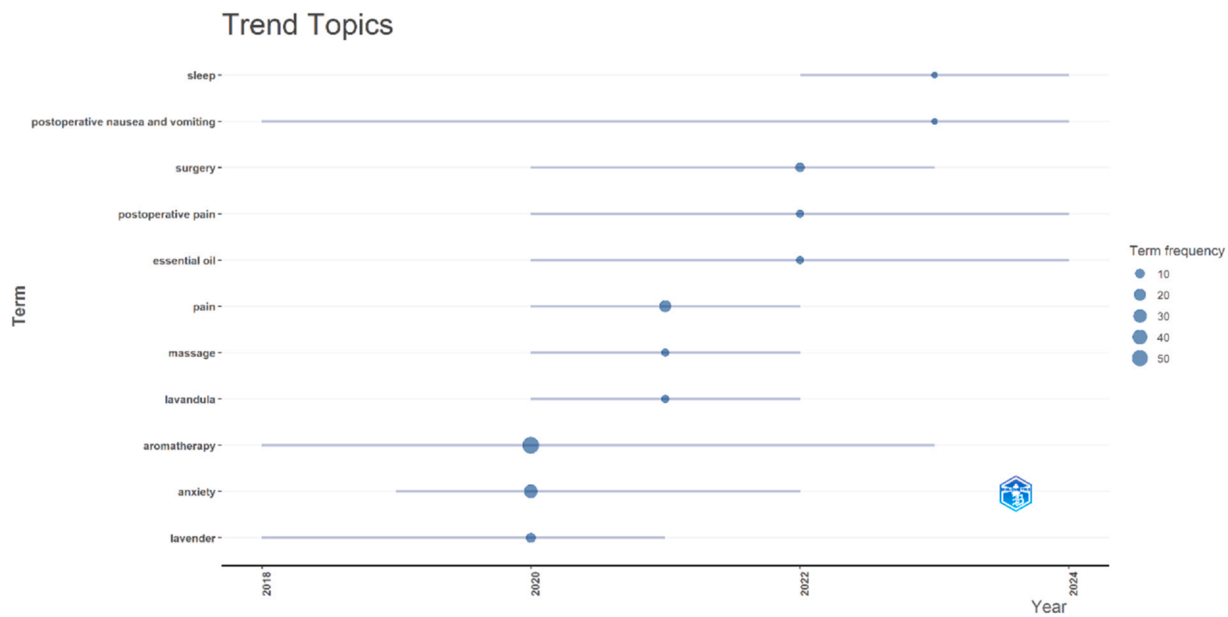

Figure 5. Trend "topics."

left quadrant includes themes with low centrality and low density, particularly those less related to the topic. Keywords here are "blind" and "lavender oil." These themes are among the emerging or declining ones. The lower right quadrant shows basic themes that have low density but high centrality. Themes located here are "aromatherapy," "depression," "essential oil," "care," and "acupressure." These topics are related to the field of research but require further development (Figure 4).

When looking at the top ten trending topics by year, a clear trend is observed toward themes such as "aromatherapy," "anxiety," "lavender," "pain," "lavandula," "massage," "surgery," "essential oil," "postoperative pain," and "postoperative nausea and vomiting" (Figure 5).

Journals Publishing Nursing Articles on Aromatherapy in Surgery

Most nursing articles related to the subject (n = 19) were published in the *Journal of Perianesthesia Nursing*. Other leading journals included *Explore—The Journal of Science and Healing* (n = 7), *Holistic Nursing Practice* (n = 6), and *Complementary Therapies in Clinical Practice* (n = 5). *Journal of Perianesthesia Nursing* has shown a significant acceleration in publications on this topic since 2017. While the journal published 16 articles on this topic until 2017, it published 16 articles in 2023 and 18 articles in 2024, totaling 34 articles in 2 years (Figure 6).

An analysis based on Bradford's Law revealed 102 journals publishing nursing articles on aromatherapy applied in surgery, classified into 3 zones. The first 4 journals were grouped in the first zone, representing the core journals most productive on the topic. According to Bradford's Law, the top four most productive journals in the first zone were *Journal of Perianesthesia Nursing* (freq = 19, cumFreq = 19), *Explore—The Journal of Science and Healing* (freq = 7, cumFreq = 26), *Holistic Nursing Practice* (freq = 6, cumFreq = 32), and *Complementary Therapies in Clinical Practice* (freq = 5, cumFreq = 37). The second zone included journals ranked from 5 to 23, while the third zone included journals ranked from 24 to 56.

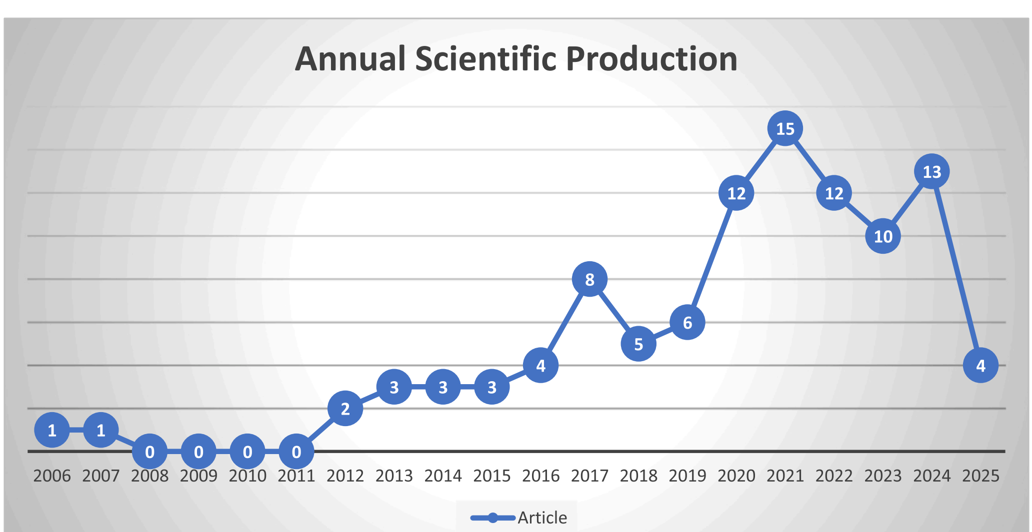

Figure 6. Annual scientific production.

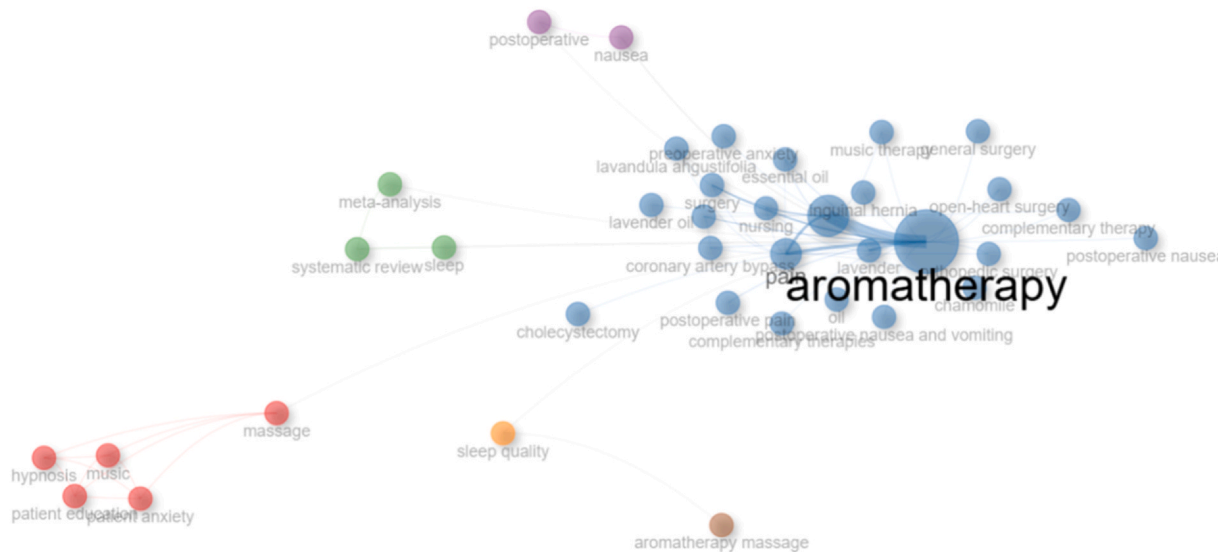

Figure 7. Co-occurrence network.

Co-author Occurrence

The frequent use of certain words in the content analysis of scientific publications indicates that these concepts are closely related, and this is made possible through co-word analysis, also known as concept mapping. In co-word analysis, the focus is not on authors, documents, or journals, but rather on the concepts being analyzed.[34,35] In Figure 7, a network analysis method was applied based on the authors' keywords. This method refers to the frequency of co-occurrence of keywords. The network analysis reveals that the keywords form six different clusters. The cluster shown in blue represents the most frequently co-occurring keywords, while the red and green clusters include frequently used keywords. In the blue cluster, the keywords "aromatherapy," "anxiety," "pain," "lavender," and "nursing" stand out in terms of usage frequency (Figure 7).

Discussion

Bibliometric studies play a significant role in mapping science and effectively conveying developments in the scientific literature. With the increasing number and scope of publications, the accessibility, analysis, and dissemination of information in academic and applied fields are becoming increasingly complex.[36] Such bibliometric and visualized studies analyze research points and international dynamics, particularly examining the trends in nursing studies on aromatherapy, providing researchers with a roadmap for exploring questions and future research directions in aromatherapy.

Within a specific discipline, the academic impact of researchers and countries is mostly assessed through the number of publications and the citations those publications receive. Citation count stands out as one of the most commonly used metrics in bibliometric analyses. Science mapping is becoming increasingly important in interdisciplinary research. With the increase in publication numbers and the deepening of specific studies, the accessibility, analysis, and transmission of knowledge in academic and applied fields are becoming more intricate processes.[37] This process can especially help nurses conducting research in areas like aromatherapy improve their practices. Bibliometric analyses in the aromatherapy literature deepen research in this field and reveal the potential of this method in nursing practice.[38,39] Studies on the effectiveness of aromatherapy, especially in postoperative care, anxiety management, and patient relaxation, contribute to the further development of nursing practices. Moreover, differences between studies conducted in various countries at an international level can help us understand how aromatherapy responds differently to global health issues.

To our knowledge, no comprehensive bibliometric analysis has been conducted in the nursing literature regarding aromatherapy in surgery. In this study, the WoS database was accessed, and the author, institution, and source networks were visualized based on publications and citations. This research is of great importance for nurses providing care to patients undergoing surgical procedures, as aromatherapy enables holistic approaches aimed at improving patient care quality by reducing anxiety and stress. Therefore, this study provides significant contributions within the context of nursing.

Publications on aromatherapy in surgery within the nursing field span the years 2006 to 2025. When examining annual scientific output, publication production on this topic was limited until 2006. The period up to 2021 shows a low number of publications, indicating fluctuating scientific interest in aromatherapy within nursing during this time. The highest publication output was recorded in 2021 (n = 15). Multidisciplinary bibliometric studies related to aromatherapy have been conducted with certain limitations.[38,39] This increase may reflect a growing demand for alternative and complementary treatment methods for surgical patients and a deeper scientific inquiry into these methods by health care professionals. However, the presence of certain limitations in multidisciplinary bibliometric studies related to aromatherapy suggests that the findings in the literature should generally be evaluated from a broader perspective. This situation indicates that the nursing literature on aromatherapy in surgery has shown increasing interest over time, although with a fluctuating trend. The limited number of publications until 2006 may indicate that aromatherapy was not yet fully explored in surgical fields and that more traditional treatment methods were preferred during this period. The fluctuation in publication numbers up to 2021 may imply that researchers in the nursing field were influenced by various factors regarding their interest in this topic. This fluctuation may be associated with the publication of studies supporting or rejecting the effectiveness of aromatherapy.

In this bibliometric analysis, Bradford's Law was used to determine the most productive journal groups in the nursing literature related to aromatherapy in surgery. Bradford's Law identifies areas of concentration in the literature by examining the distribution of scientific publications across journals.[40] According to this law, journals are divided into three main zones, each containing a similar number of articles. The first zone includes the core journals with the most publications related to the field; the second zone contains journals that, while less intensive, still significantly contribute to the literature; and the third zone typically consists of journals with fewer publications that focus on more specific areas.[41] Bradford's Law offers an objective method for evaluating the scientific contribution level of journals. Additionally, bibliometric indicators such as literature aging rate, half-life, and impact factor provide further information on the impact level and scientific value of journal publications.[42]

According to the Bradford-based analysis, the top four most productive journals in the nursing literature on aromatherapy in surgery are the *Journal of Perianesthesia Nursing*, Explore: *The Journal of Science and Healing*, *Holistic Nursing Practice*, and *Complementary Therapies in Clinical Practice*. A total of 102 journals classified into three zones according to Bradford's Law were identified as publishing nursing articles on aromatherapy in surgery. The fact that these publications are found in journals specific to aromatherapy provides an opportunity to reach the target audience of nurses. This contributes to the professional development of nursing care in the implementation of aromatherapy. Furthermore, these journals can be considered core journals for the dissemination of knowledge in the field of aromatherapy in nursing.

Keywords provide insight into the research topics mentioned by the authors in their studies. Keywords make an important contribution to determining the focus areas within a research field. A bibliometric study on aromatherapy focused on topics such as anxiety, depression, and insomnia.[38] Another bibliometric analysis of aromatherapy concentrated on essential oils, intervention, and complementary medicine.[39] The keywords for this study were determined as aromatherapy, anxiety, pain, lavender, nursing, and surgery. Patients undergoing surgical procedures often experience preoperative anxiety.[14] In a meta-analysis by Kulakaç and colleagues investigating the effect of lavender oil on preoperative anxiety, it was revealed that the application of lavender oil before surgery was an effective method of reducing patients' anxiety.[43] Stanley et al[12] applied lavender oil to patients undergoing cataract surgery for preoperative anxiety and found that lavender aromatherapy reduced preoperative anxiety.[12] In a study by Mosamosseini et al. (2023) evaluating the effect of aromatherapy on pain in patients undergoing coronary bypass surgery, aromatherapy massage significantly reduced pain intensity in patients.[22] In a study assessing the effect of lavender oil on reducing postoperative shoulder pain in patients undergoing laparoscopic cholecystectomy, lavender oil was found to be effective in pain reduction.[25] In contrast to the findings of this study, a meta-analysis investigating the effects of aromatherapy on anxiety demonstrated that aromatherapy with Citrus aurantium, Bergamot and Citrus sinensis, rose oil, grapefruit, and Roman chamomile also had effects on anxiety[8,21] emphasized that the use of Damascus rose and chamomile aromatherapy in emergency orthopedic surgery patients effectively reduced anxiety and pain levels.[21] A study evaluating the effect of inhaled geranium oil aromatherapy on anxiety in patients undergoing hernia surgery reported that geranium aromatherapy was effective in reducing anxiety levels.[6] Examples in the literature indicate that, in addition to lavender oil, various essential oils are being applied in nursing and are gaining popularity, particularly for anxiety and pain management, suggesting that they may become even more popular in the coming years. The findings of this research and the examples in the literature show that lavender oil is applied in various ways in surgical aromatherapy practices conducted in the nursing field and that it is gaining popularity, particularly for anxiety and pain management, suggesting it may become even more popular in the coming years.

According to the results of the thematic map, the most frequently researched topics are "pain," "health," and "mood." Niche themes, expressed as high-density and low centrality, can delve deeply into a specific interest or research area but generally do not attract as broad an audience as more popular topics. In this study, the niche themes were "prevention," "cancer," and "cognitive-behavioral therapy." It is recommended that researchers and academics focus more on these under-researched themes to contribute to the development of more in-depth literature in this field.

This study identified the most prolific author as being from Iran. The use of herbal oils in Iran has a deep-rooted place in traditional Iranian medicine, dating back over 3,000 years. The inheritance of aromatherapy from Eastern medicine, the acceptance of traditional herbal treatments, and the positive attitudes of health care professionals are among the factors influencing the preference for aromatherapy.[44] However, a bibliometric analysis of aromatherapy for anxiety, depression, and insomnia reported that the most prolific country was the United States.[38] This difference is thought to be due to the high prevalence of anxiety and insomnia in the United States.

Strengths and Limitations

The important strength of this study is that it is the first comprehensive bibliometric analysis focusing on nursing publications on aromatherapy in surgery. Another strength is that it shows the common and different characteristics of nursing and multidisciplinary literature on the subject. Another strength is that the literature was selected from WoS, one of the most reliable databases. The study also has certain limitations. Nursing studies in this field are also published in journals not indexed in WOS. Therefore, the primary limitation of the study was that choosing another database could have yielded different results. Second, the analysis included only articles and review articles. Including different publication types such as book chapters, conference proceedings, and others may lead to different results.

Conclusion

This bibliometric analysis revealed that the literature on aromatherapy in surgical applications in the field of nursing showed an increasing publication trend between 2006 and 2025. The most prolific author was found to be from Iran, and the journal with the most publications was the *Journal of Perianesthesia Nursing*. The most frequently used terms in the keyword analysis were "aromatherapy," "anxiety," "pain," "lavender," "nursing," and "surgery." Thematic map analysis showed that topics such as "pain," "health," and "mood" were among the driving themes, while topics such as "aromatherapy," "depression," and "essential oil" needed to be further developed as core themes. Trend analysis by year showed that topics such as "aromatherapy," "anxiety," "lavender," and "postoperative pain" stood out. Co-author network analysis revealed that the concepts of "aromatherapy," "anxiety," "pain," "lavender," and "nursing" were most frequently used together. The findings indicate that research on aromatherapy in the field of surgical nursing is increasing, particularly highlighting the role of lavender oil and essential oils in anxiety and pain management, and suggesting that these topics require more comprehensive studies in the future.

CRediT Authorship Contribution Statement

Çağla Toprak: Writing – original draft, Resources, Formal analysis, Conceptualization. **Seda Cansu Yeniğün Akbulut:** Writing – review and editing, Writing – original draft, Resources, Methodology, Conceptualization.

Data Availability

All data generated or analyzed during this study are included in this published article. For other data, these may be requested through the corresponding author.

Declaration of Competing Interest

None to Report.

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